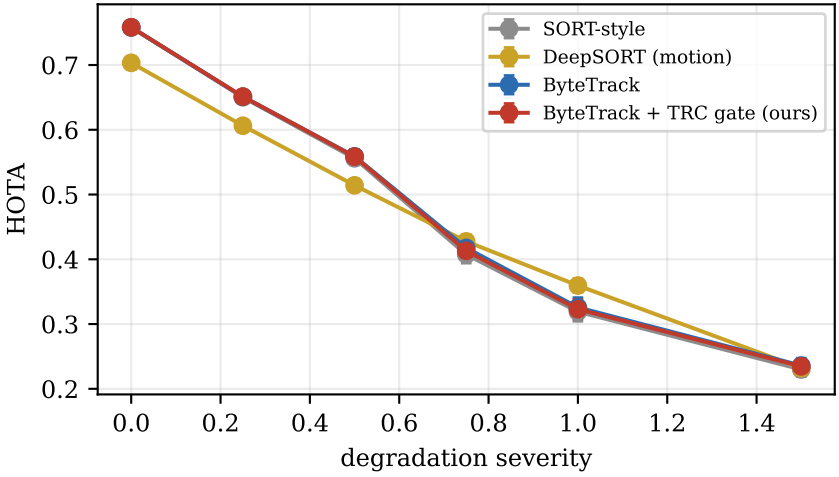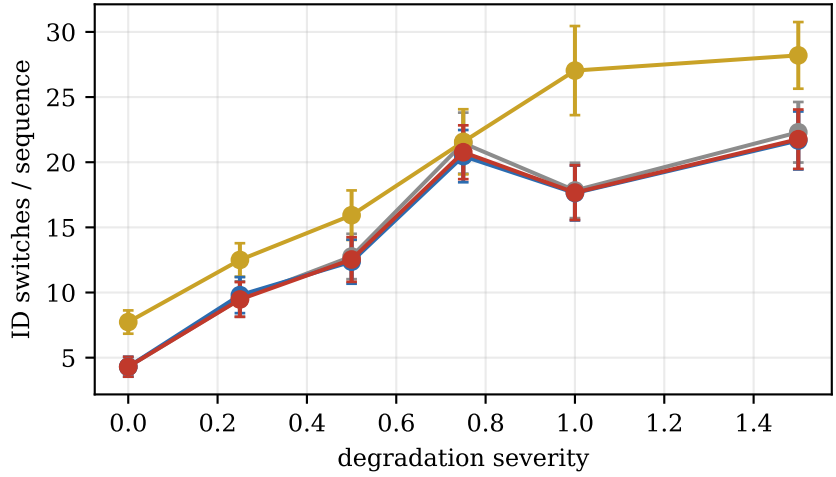Synthetic benchmark: robustness to detector degradation (error bars = s.e.m.)